Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. We trained each model for 50 epochs with a batch size of 128.

2 Results

Figure 1 shows the accuracy by dataset.

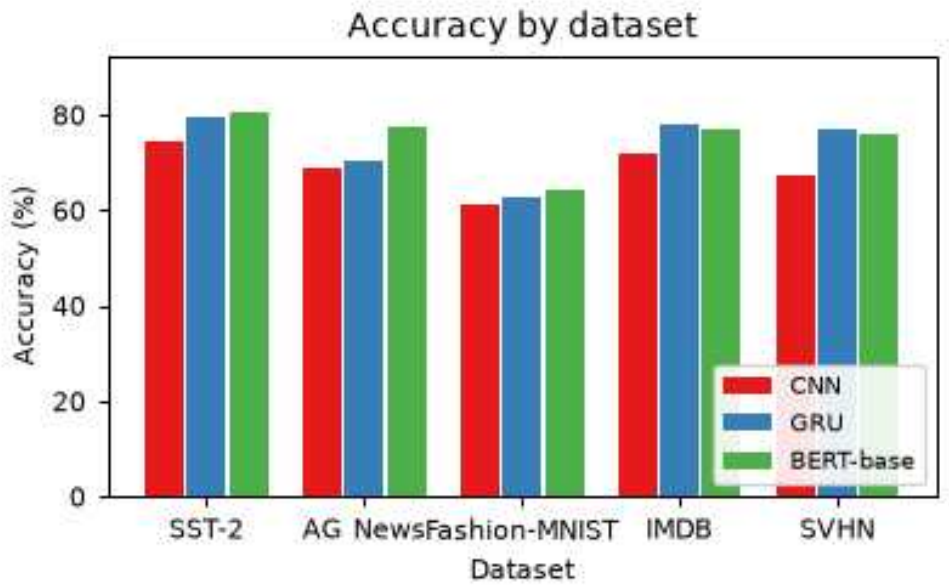


Figure 1: Accuracy by dataset for the models in this experiment.

BERT-base outperforms CNN by 8.5 percentage points on AG News. On SST-2, GRU obtains a accuracy of 75.1%.

Figure 2 shows the test accuracy of CNN by dataset.

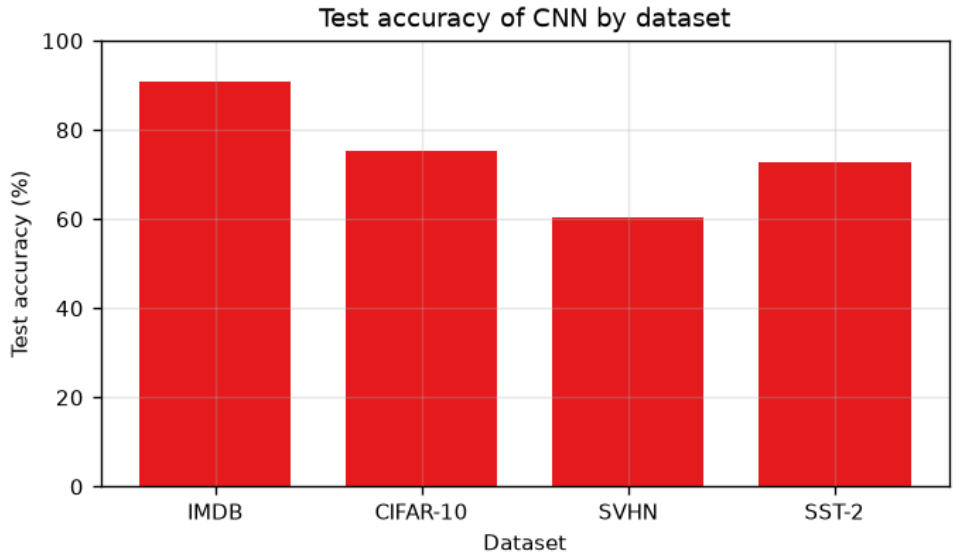


Figure 2: Test accuracy of CNN by dataset for the models in this experiment.

CNN achieves 60.3% test accuracy on SVHN. CNN reaches over 85% test accuracy on IMDB (Figure 2).

Figure 3 shows the test accuracy of LSTM by dataset.

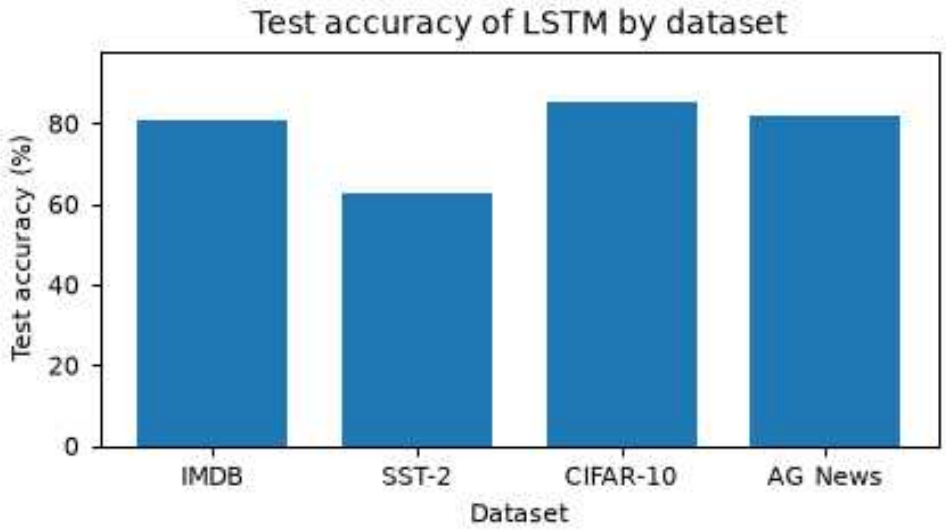


Figure 3: Test accuracy of LSTM by dataset for the models in this experiment.

LSTM achieves 62.5% test accuracy on CIFAR-10. On SST-2, LSTM obtains a test accuracy of 72.3% (Figure 3). LSTM achieves 83.9% test accuracy on AG News (Figure 3). LSTM reaches 70.9% test accuracy on IMDB (Figure 3).

3 Discussion

Overall, the results are in line with what we expected from the literature.